# DATA ANALYST INTERNSHIP — TASK 1

# Data Cleaning and Preprocessing

**Dataset:** Medical Appointment No Shows (Kaggle)

**Tool:** Python and Pandas

## 1. Objective

To inspect a raw medical appointment dataset and prepare a consistent dataset for further analysis by checking missing values, duplicates, inconsistent text, date formats, and invalid records.

## 2. Cleaning operations

| Area | Action performed |
|---|---|
| Column names | Converted column names to lowercase with underscores and corrected known spelling errors. |
| Duplicates | Checked and removed exact duplicate rows. |
| Missing values | Inspected missing values; age is filled with the median and gender with the mode only if missing values are present. |
| Text values | Standardized gender, neighbourhood, and No-show labels. |
| Dates | Parsed ScheduledDay and AppointmentDay as UTC datetime values. |
| Data types | Converted age and indicator fields to numeric types; kept IDs as strings. |
| Invalid ages | Removed ages below 0 or above 120; retained age 0. |
| Waiting time | Created wait_days and removed records where appointment date precedes scheduling date. |
| Handicap field | Preserved the source numeric categories rather than assuming only 0 and 1. |

## 3. Output files

The cleaning program generates medical_appointments_cleaned.csv and cleaning_summary.txt in data/cleaned/. The numeric counts are generated from the actual source file at run time and should be copied from that log into the submission.

## 4. Conclusion

The preprocessing workflow creates a more consistent dataset suitable for subsequent analysis. It preserves the meaning of the original No-show field: Yes means the patient did not attend, while No means the patient attended.

Dataset source: Kaggle — https://www.kaggle.com/datasets/joniarroba/noshowappointments

Note: The raw Kaggle CSV is not included in this package. Download it from the source and place it in data/raw/ before running the notebook or script.